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Interactive Computer Graphics

Chapter 4 – Geometric Projection

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Projecting 3D Models

- To compute the 2D representation of a 3D model on the screen, it needs to be projected, whereby it loses its z-coordinates
- Depending on the projection type, the object might get skewed (perspective shortening)
- Projection can also be expressed as a matrix multiplication, as done with geometric transformations

Chapter Outline

4.1 Linear Perspective

4.2 Planar Projections

4.3 Camera Modeling

4.4 Specifying Projections in OpenGL

4.5 Perspective Projection Derivation

4.6 Orthographic Projection Derivation

4.7 Viewport Transformation

References



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4.1 Linear Perspective

Discovery of perspective distortion in art



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Linear Perspective

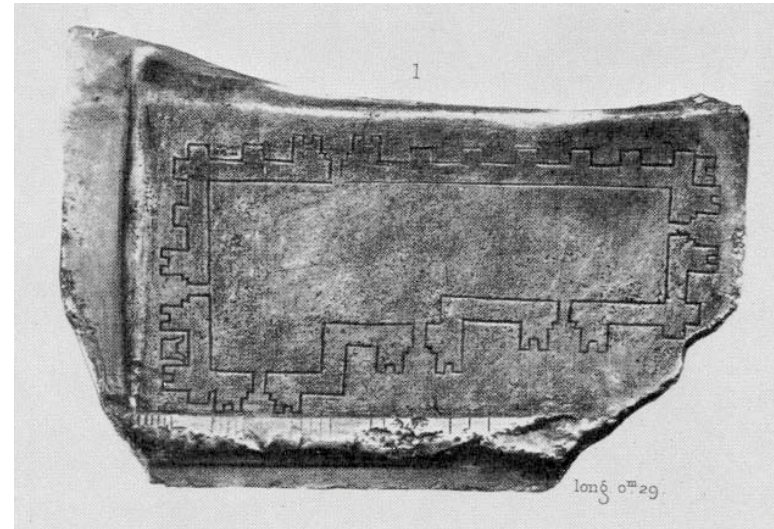
- What is wrong in this image?

Satire on False Perspective
William Hogarth (1753)



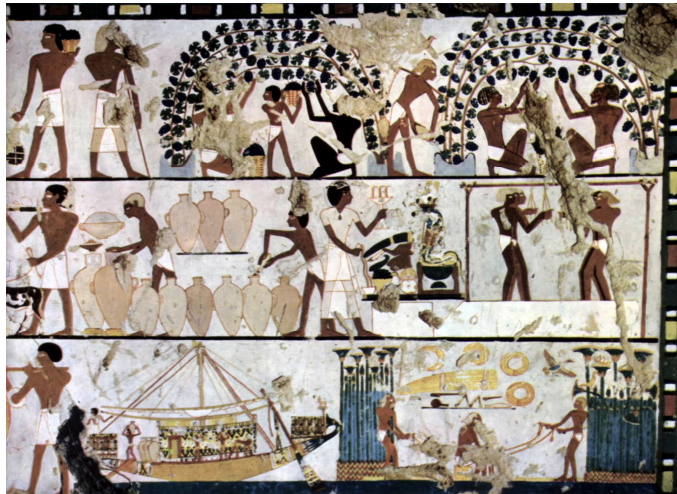
Drawing by Projecting

- Projection: method for imaging objects using parallel or central beams
- Perspective:
"The science which teaches how to represent tri-dimensional objects on a bi-dimensional surface, so that the perspective image coincides with the one which is given by direct vision."
(Reina at La Prospettiva, Milan, 1940)
- Early example of a projection: ground plan of a Mesopotamian building 2150 BC
(one of the first known technical drawings)



Projection in the Arts 1/5

- Before the Renaissance, pictures often had a religious and symbolic character
- Perspective was not understood: accordingly, they seem 2-dimensional



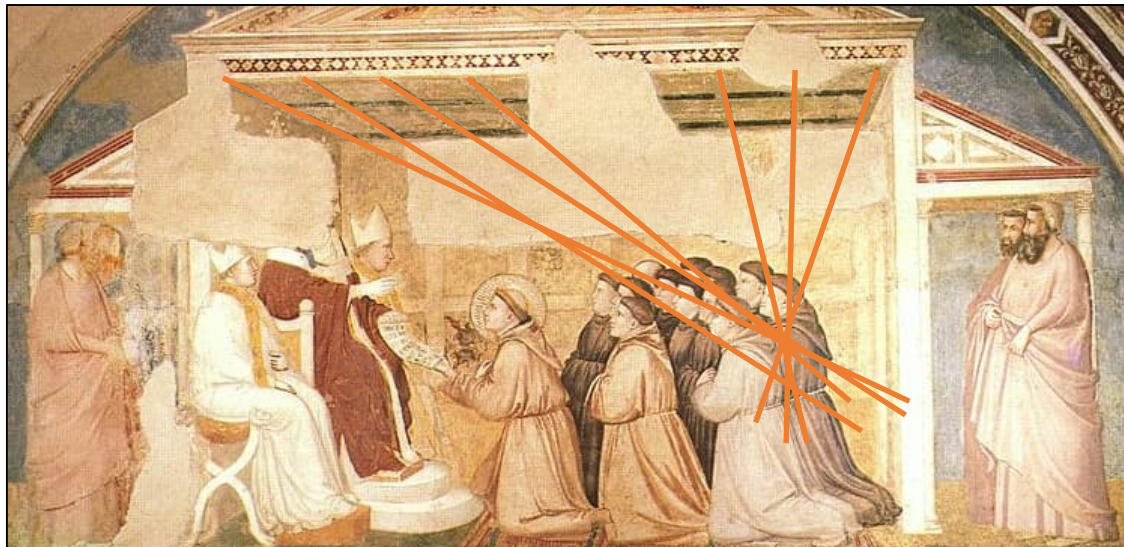
Fresco from an Egyptian grave
(circa 1500 BC)

Madonna and Child on a Curved Throne,
Byzantine School (circa 1325)



Projection in the Arts 2/5

- Techniques to simulate space:
 - Shaded round volumetric shapes
 - Spatial depth through converging lines



Line convergence is not systematic, lines do not converge in a single vanishing point

Giotto di Bondone
(1266-1337, Florence)
Confirmation of St. Francis
(circa 1325)

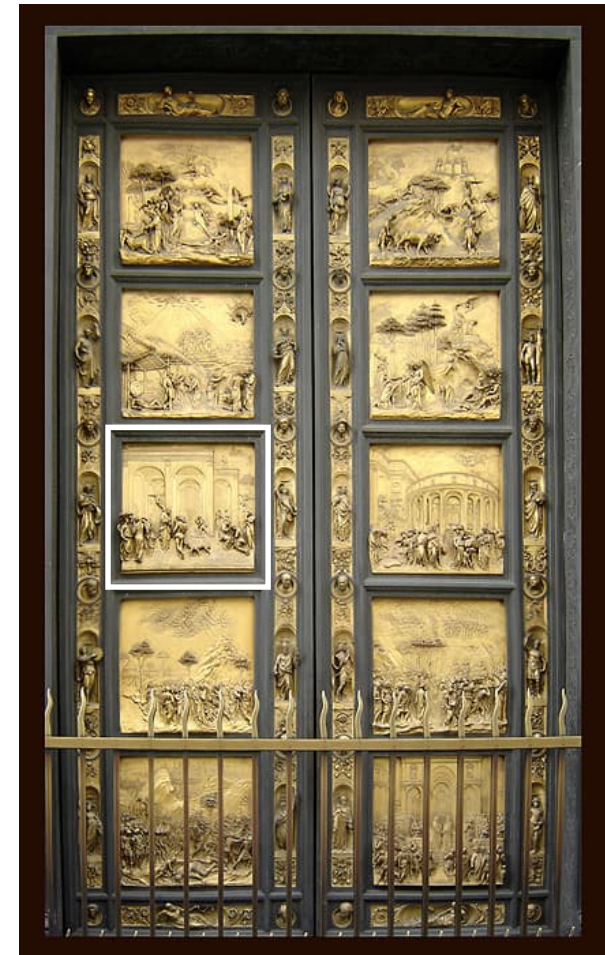
Projection in the Arts 3/5

- Renaissance (1400-1700)
 - New emphasis on the importance of personal views and interpretation of the world, power of observation, especially of nature (astronomy, anatomy, botany, ...)
 - Universe as a clockwork: intellectual reconstruction of the universe according to mechanical principles

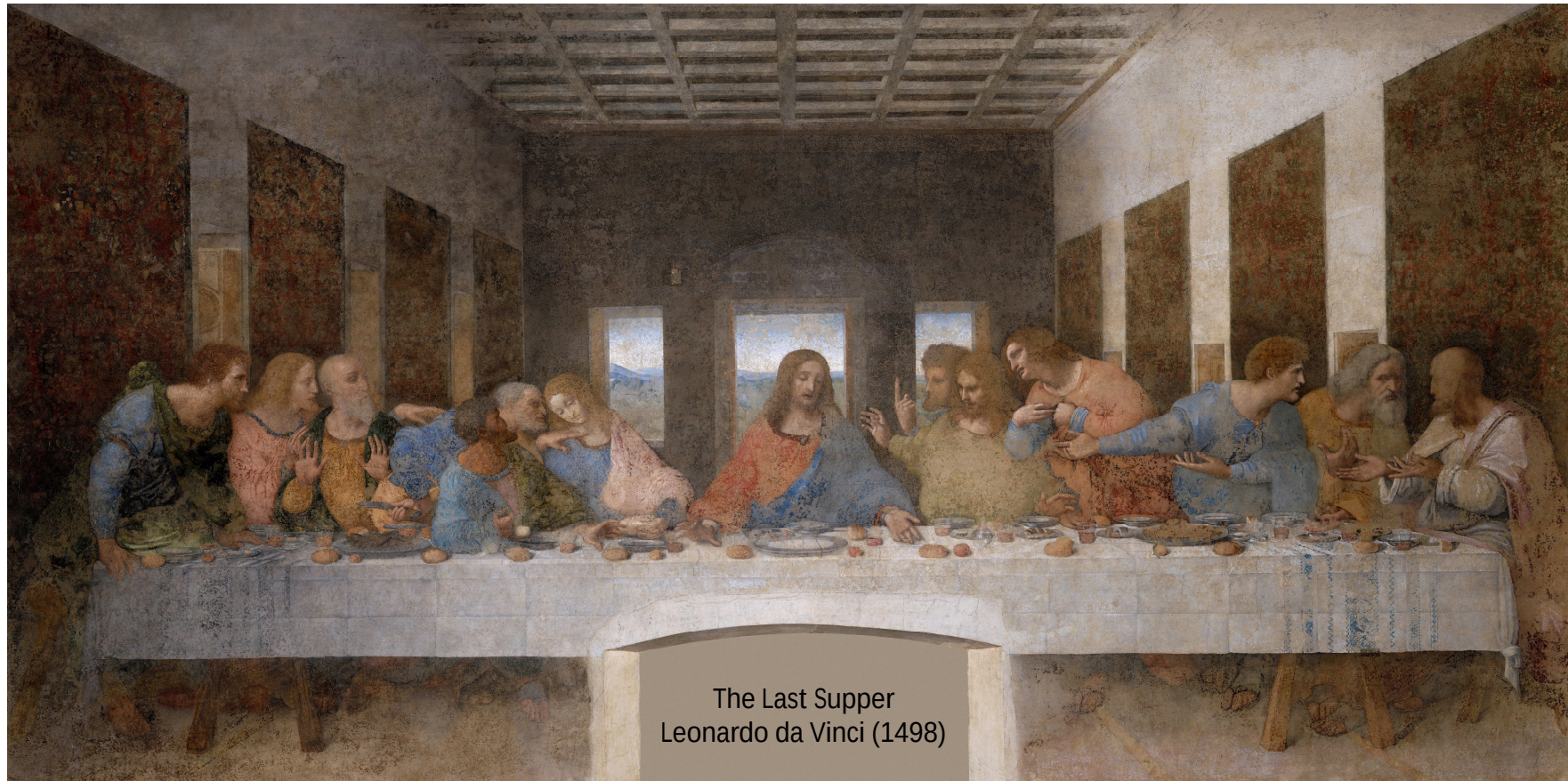


Projection in the Arts 4/5

- Examples from the Renaissance



Projection in the Arts 4/5



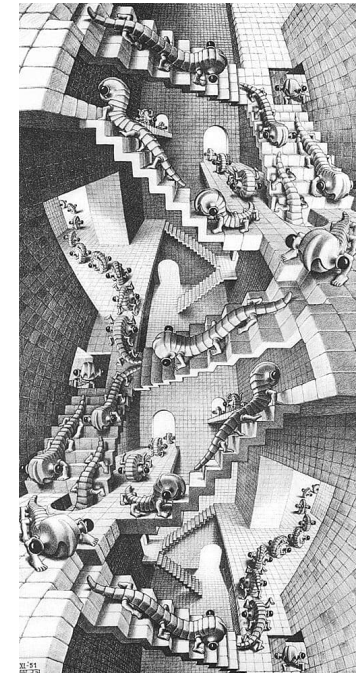
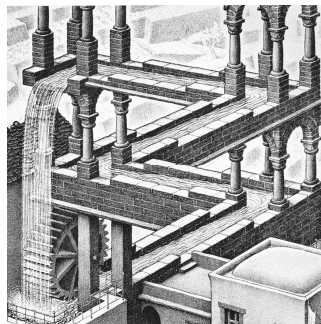
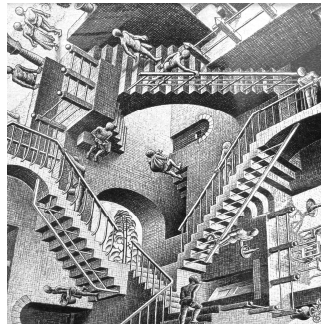
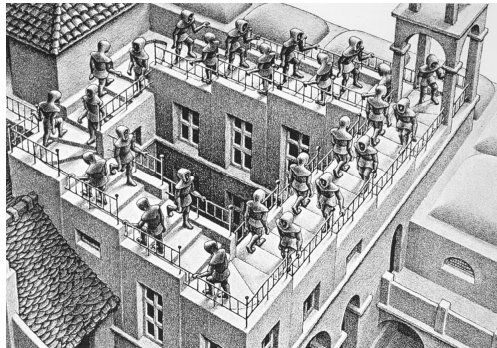
Projection in the Arts 5/5

- "The Invention of the Painting" (1830) by Karl Friedrich Schinkel (1781-1841):
 - An image based on a legend by Pliny the Elder
 - Tracing the shadow contours on a drawing surface



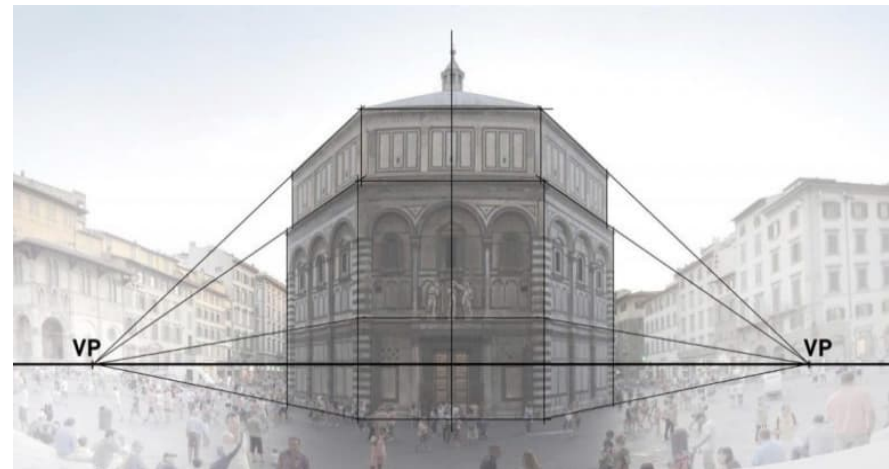
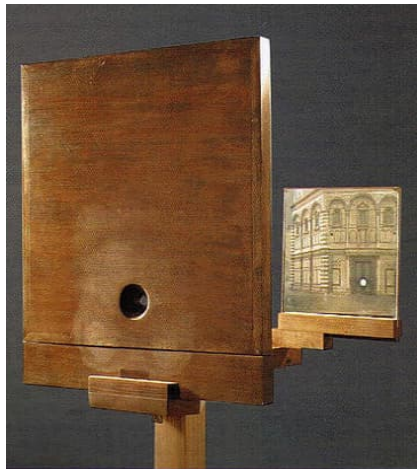
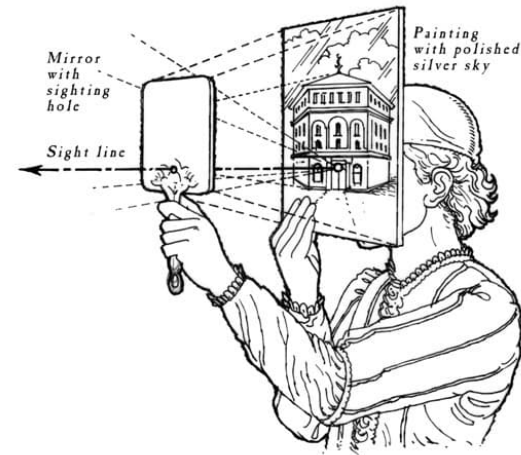
Projection in the Arts 6/5

- M. C. Escher (1898 – 1972)



Projection Technics 1/3

- Filippo Brunelleschi (1377-1446):
Systematic method for obtaining
a perspective projection



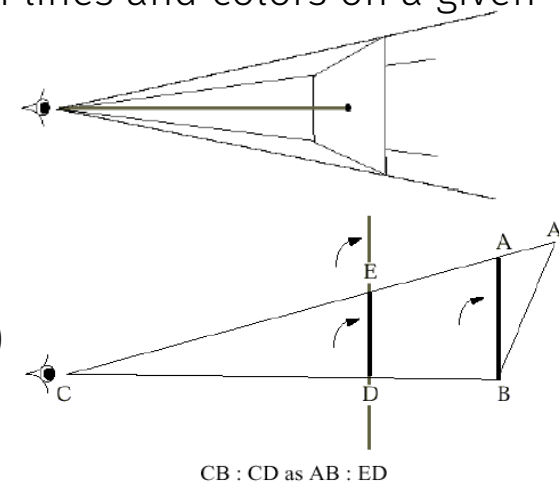
Projection Technics 1/3



The Delivery of the Keys
Perugino (1482)

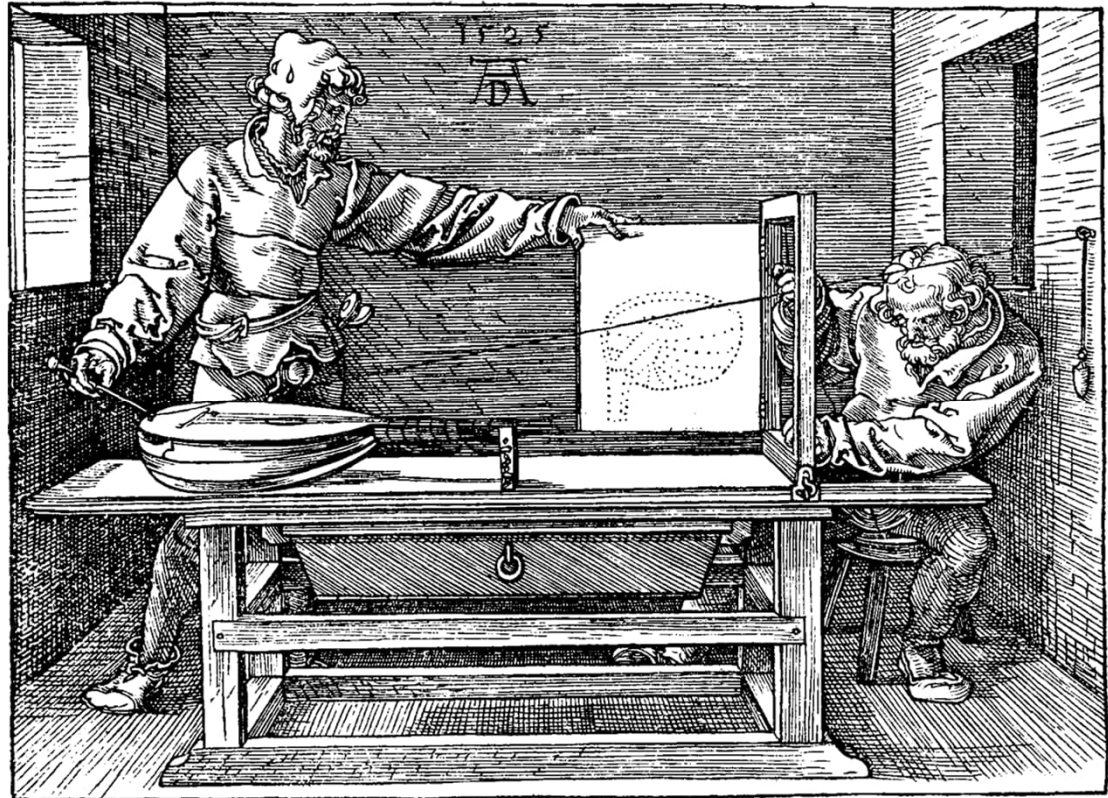
Projection Technics 2/3

- Formal description of the perspective projection
 - Leone Batista Alberti (1404-1472):
First essay on the subject of "La Pittura" (1435)
 - "A painting [the projection plane] is the intersection of a visual pyramid [view frustum] at a given distance, with a fixed center [center of projection] and a defined position of light, represented by art with lines and colors on a given surface [the rendering]."
(Alberti, On Painting)
- Determination of the projected image:
 - Height of the object (AB)
 - Distance from eye to object (CB)
 - Distance from the eye to the image plane (CD)
 - Similar triangles: $CB:CD$ behaves like $AB:ED$ (intercept theorem)



Projection Technics 3/3

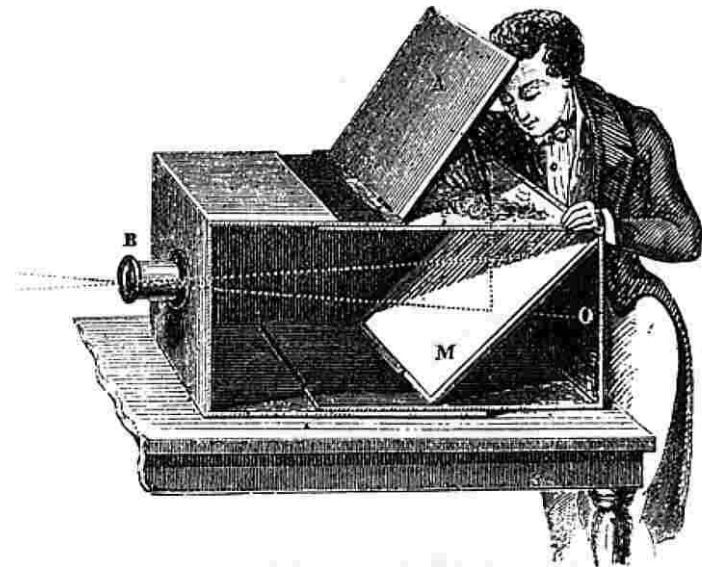
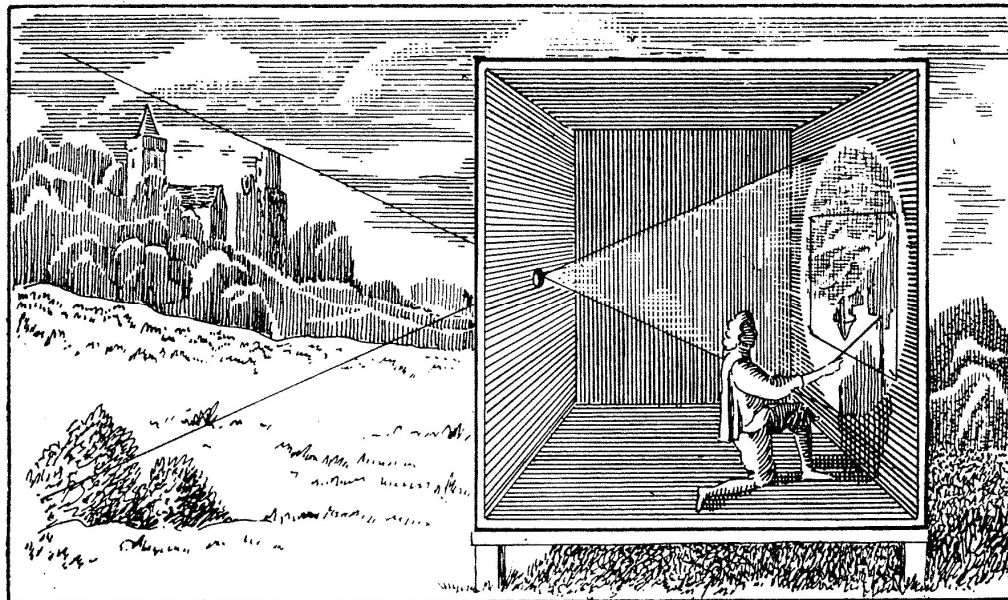
- Albrecht Dürer (1471-1528): Geometric and mechanical description of the concept of similar triangles
- Construction of several "projection apparatuses"



Der Zeichner der Laute
Albrecht Dürer (1525)

Projection Technics 3/3

- Camera Obscura





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4.2 Planar Projections

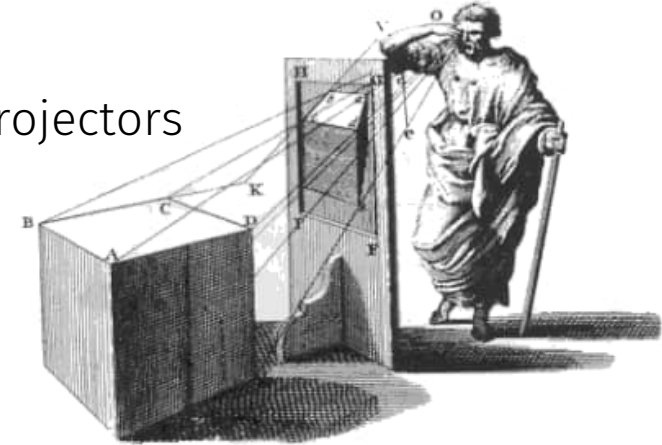
Perspective and parallel projections



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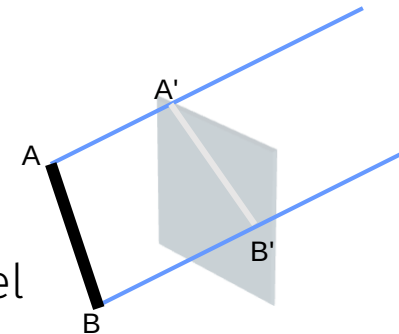
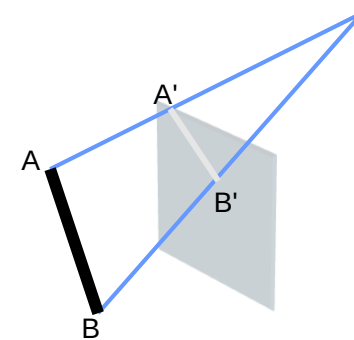
Geometric Projections

- Method of mapping objects onto a screen using projectors
- Planar geometric projections
 - Projection beams are straight lines
 - Projection surface is a plane (image plane, picture plane, view plane, projection plane, ...)
- Non-planar geometric projections:
 - *Curved* projection beams and projection surfaces (e.g., fisheye lens)

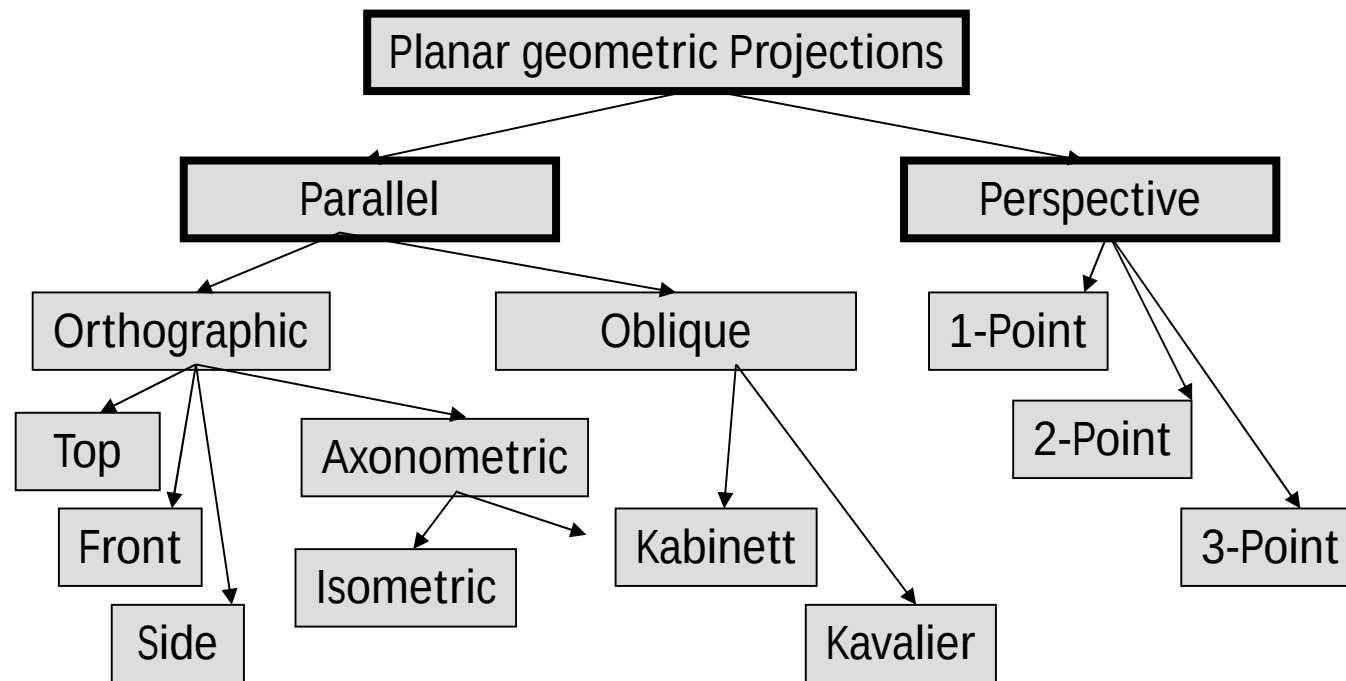


Projection Types 1/2

- Perspective projection (= central projection)
 - Established by projection center (COP, center of projection)
 - Perspective reduction: The size of the projection of an object decreases with increasing distance from the projection center
- Parallel projection
 - Determined by projection direction (DOP, direction of projection)
 - Receives sizes of the objects projected in parallel

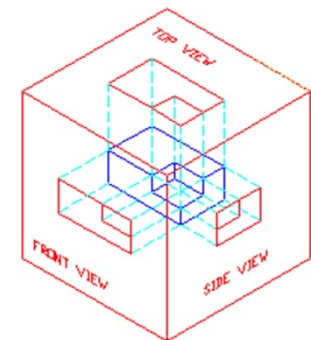
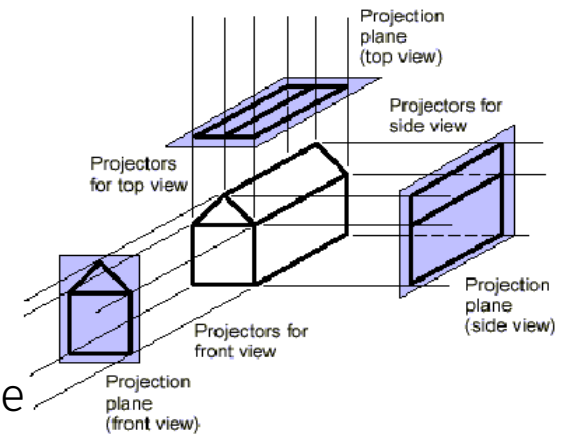


Projection Types 2/2



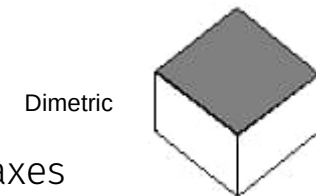
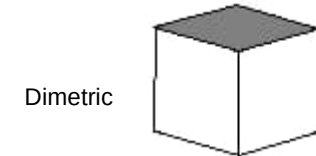
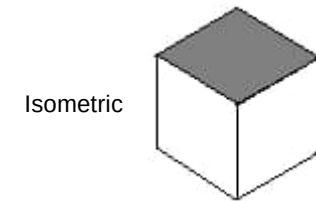
Orthographic Projections

- Orthographic projections often used for construction drawings of machines and parts
- Main views: Front view, top view and side view
- Advantages
 - Accurate length and angle measurements possible
 - all views use the same scale
- Disadvantage
 - No realistic representation for 3D objects and their environment
 - Multiple views necessary to gain spatial idea of a 3D object



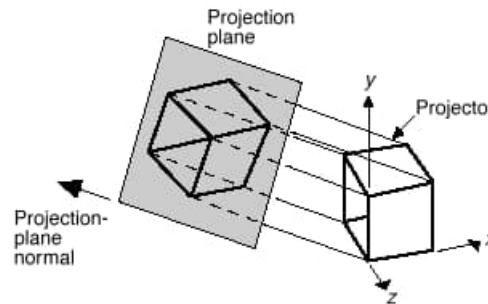
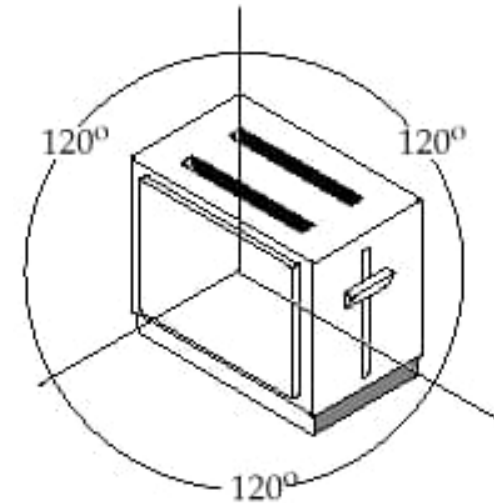
Axonometric Projections

- Projection beams perpendicular to the projection plane
- Projection plane not perpendicular to a coordinate axis
- Visible effects
 - Several perpendicular surfaces visible
 - Parallel lines are shortened equally
- Isometric
 - Normal of projection plane forms same angle with *all* main axes
 - All major axes are shortened in same ratio and relative proportions are preserved
- Dimetric
 - Normal of projection plane forms same angle with *two* main axes
 - Same shortening ratios for two main axes
- Trimetric
 - Normal of projection plane forms different angles with the coordinate axes
 - Any shortening conditions along the main axes possible



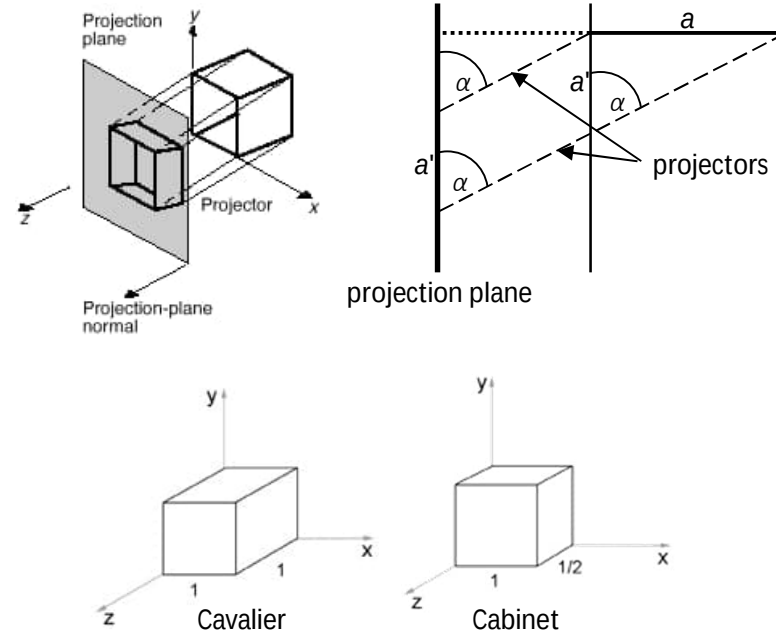
Isometric Projections

- Applications in illustrations, furniture design, ...
- Advantages
 - Communicate spatial impression in a single view
 - Same length ratios along the main axes
- Disadvantages
 - Missing shortening creates a distorted appearance
 - Less useful for objects with curved surfaces



Oblique Projection 1/2

- Projection rays not perpendicular to the projection plane
 - Common: $\alpha = 30^\circ$
- Cavalier projection
 - $\alpha = 45^\circ$, i.e., $\tan(\alpha) = 1$
 - Lengths on straight lines perpendicular to the projection plane remain unchanged
- Cabinet projection
 - $\alpha = 63.4^\circ$, i.e., $\tan(\alpha) = 2$
 - Lengths on straight lines perpendicular to the projection plane are halved



Oblique Projection 2/2

- Advantages

- Can accurately represent the shape of a boundary surface of an object for accurate measurements
- Missing foreshortening simplifies comparison of sizes
- Gives a good impression of the 3D appearance

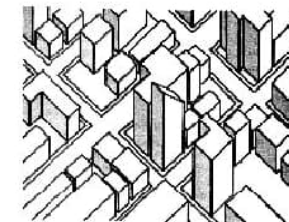
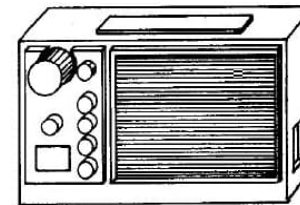
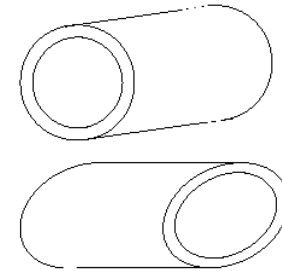
- Disadvantage

- Objects appear distorted if the projection plane is not carefully chosen (e.g., circles to ellipses)
- Missing reduction (no realistic appearance)

- Choice of projection level

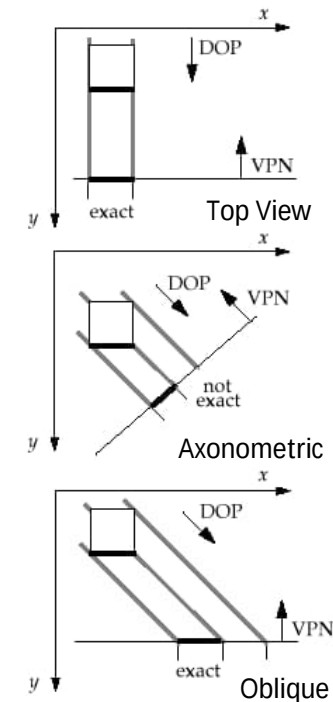
- Parallel to the most irregular main boundary area of the object to be displayed
- Parallel to the boundary plane containing circular or curved surfaces
- Parallel to the longest main boundary surface of the object
- Parallel to the boundary surface, which is of the greatest interest

- Examples: technical illustrations (radio, city view)



Properties of Parallel Projections

- Assumption: Object surface of greatest interest lies parallel to a main plane
 - Direction of projection: DOP (direction of projection)
 - Normal of the projection plane: VPN (view plane normal)
- Top view (orthographic)
 - $\text{VPN} \parallel \text{Main axis}$, $\text{DOP} \parallel \text{VPN}$
 - Shows single object surface and allows exact measurements
- Axonometric
 - VPN not $\parallel$ to main axis, $\text{DOP} \parallel \text{VPN}$
 - Adjacent areas, none exact, are visible in the picture; even shortening (as a function of the angle between surface normal and DOP)
- Oblique
 - $\text{VPN} \parallel \text{Main axis}$, DOP and VPN not parallel
 - Adjacent surfaces, one exact, the other shortened, are visible in the figure



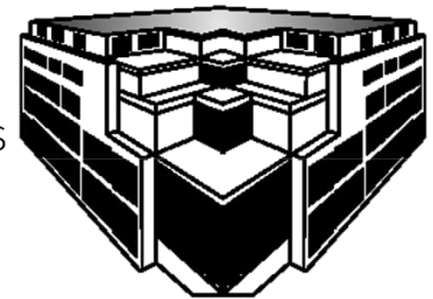
Perspective Projections 1/3

■ Properties

- Communicate realistic views of three-dimensional objects and give a spatial impression
- Distort objects in perspective view
- Scaling of objects is lost (except where the object intersects the projection plane)

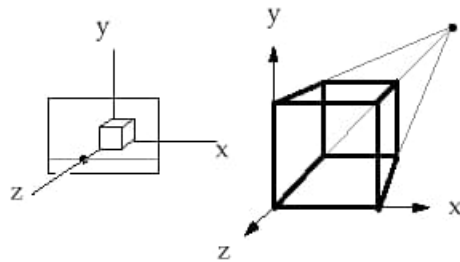
■ Differences to parallel projection

- Parallel straight lines that are not parallel to the projection plane are displayed in intersecting straight lines
- The size of an object decreases with increasing distance to the projection center
- Shortening is not uniform

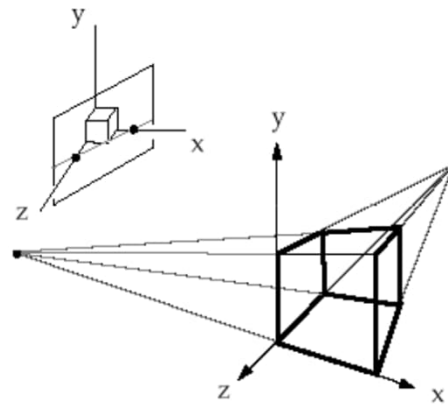


Perspective Projections 2/3

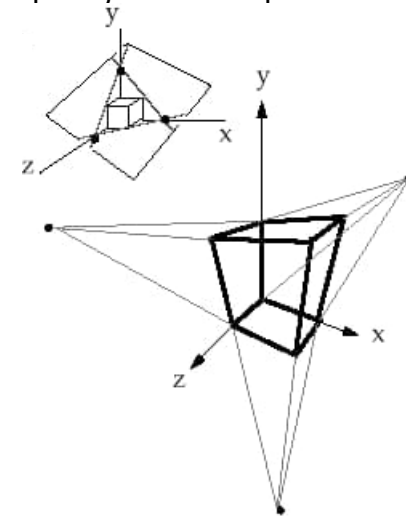
- Vanishing points
 - For rectangular objects (e.g., axis parallel cubes) whose surface normals are parallel to the x-, y-, or z-axis, the number of vanishing points is equal to the number of coordinate axes intersected by the projection plane



1-Point Perspective



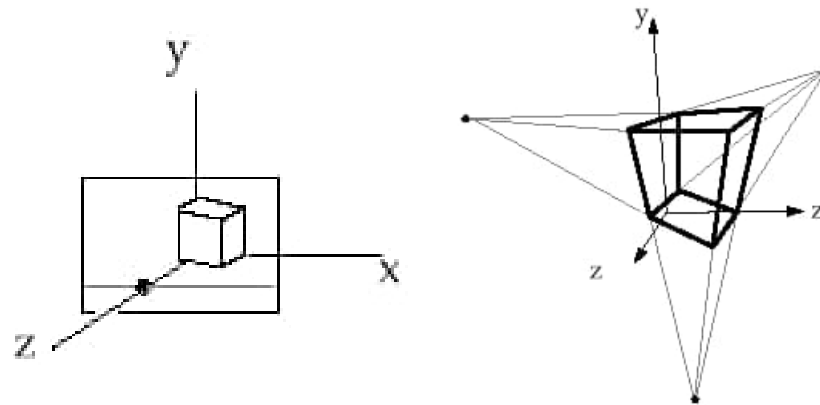
2-Point Perspective



3-Point Perspective

Perspective Projections 3/3

- What happens if the cube is rotated so that its surface normals are not parallel to the coordinate axes?
- Although the projection plane intersects only one coordinate axis, in this case you get 3 vanishing points





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4.3 Camera Modeling

Orientation of the camera and definition of the view volume



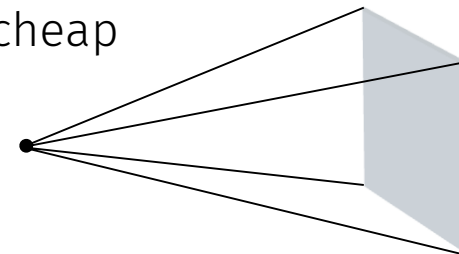
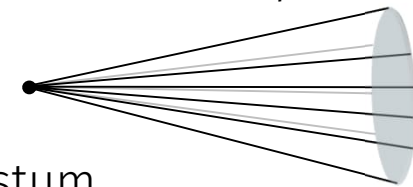
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Camera Modeling

- Application of planar geometric projections, represented by 4×4 matrices, to project a 3D scene onto a projection screen, i.e., the canvas
- Basic camera parameters
 - Camera position
 - Camera orientation
 - Projection type (parallel or perspective)
 - Angle of view
 - Depth restriction (far distance, near distance)
- Disregarded parameters
 - Distance of the image plane from the origin (focal length)
 - Depth of field
 - Inclination of the film plane (tilt of view plane)

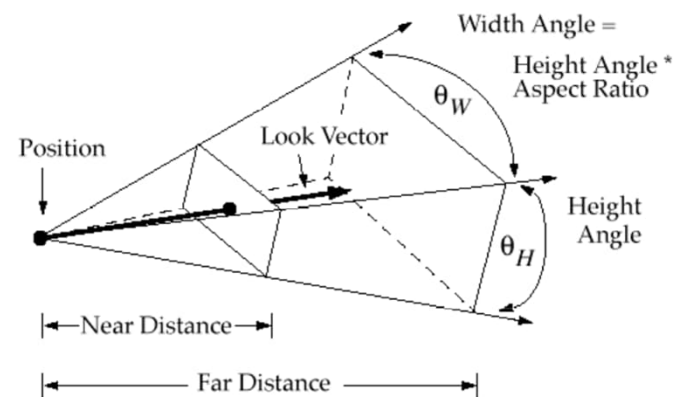
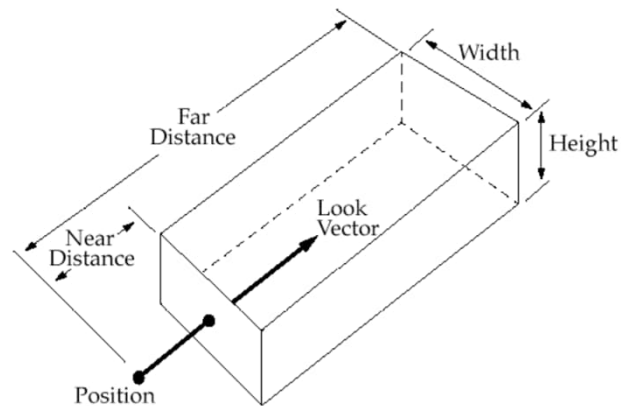
View Volume

- Objects that are within the view volume are visible for the camera
- Conical viewing volume
 - Clipping the scene objects against a conical viewing volume is expensive (solving quadratic equations)
- Frustum - rectangular viewing volume
 - Approximation of the conical visual volume by a frustum
 - Fits to a rectangular window
 - Clipping the scene objects against a frustration is cheap (solving linear equations)



Projection Specification

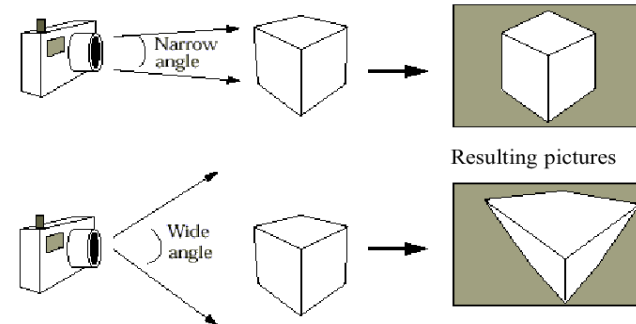
- Specified by Height Angle, Aspect Ratio, Near Plane Distance and Far Plane Distance



- Examples of Aspect Ratio
 - NTSC 4 : 3, HDTV 16 : 9
 - Cinema 2.35 : 1, 2.39 : 1, 2.40 : 1, or 2.55 : 1
 - IMAX Leonberg 38.80 m : 21.00 m

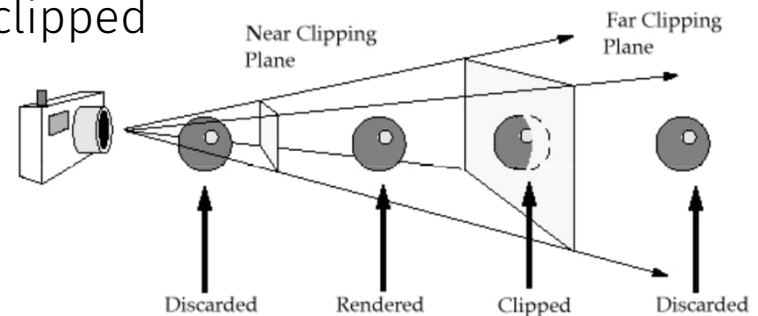
View Angle

- Parallel projection: no viewing angle as rays are parallel
- Perspective projection:
viewing angle determines perspective distortion in the image
- Specification
 - Through the viewing angle in height (field of view in y, fovy)
 - The viewing angle in the width results from the viewing angle in the height multiplied by the aspect ratio: $\text{width angle} = \text{height angle} \cdot \text{aspect ratio}$
 - Strong distortion in a wide-angle lens



Clipping Planes

- Position of the planes is determined by the distance along the look vector:
near distance and far distance
 - Near plane hides objects that are behind or too close to the camera
 - Far plane hides objects that are too far away (and may be hard to spot)
⇒ Reduction of rendering overhead
- Clipping
 - Objects outside the view volume are not drawn
 - Viewing volume intersecting objects are clipped





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4.4 Specifying Projections in OpenGL

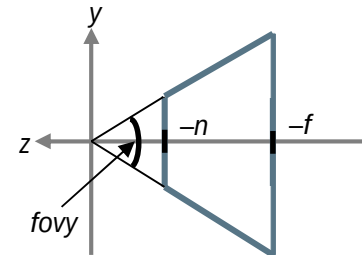
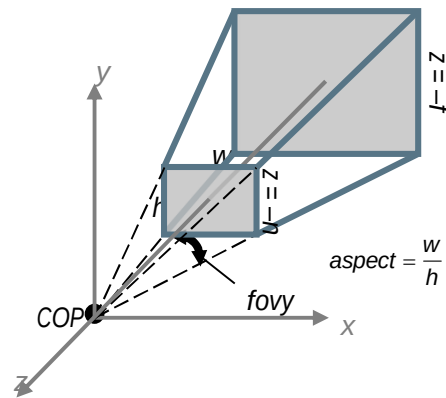
Functions to describe the view volume



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Projection Matrix Generation 1/3

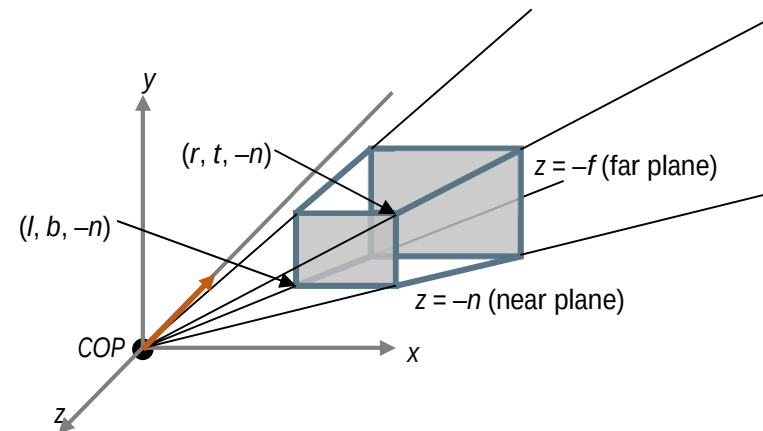
- Convenient generation of perspective projection matrix with GLM
- `glm::perspective(fovy, aspect, near, far)`



- View volume is symmetrical with respect to the z-axis
- Coordinate system still right-handed

Projection Matrix Generation 2/3

- Alternative generation of perspective projection matrix with GLM
- `glm::frustum(left, right, bottom, top, near, far)`



- View volume does not have to be symmetrical with respect to the z -axis
- Coordinate system still right-handed

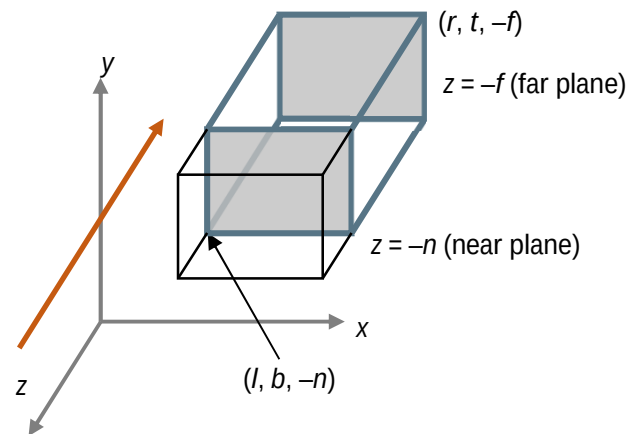
Projection Matrix Generation 3/3

- Implementation of glm::perspective

```
detail::tmat4x4<T> perspective(T const & fovy, T const & aspect,  
                               T const & near, T const & far) {  
    T const & l, r, b, t;  
    t = n * tan(fovy * M_PI / 360.0);  
    b = -t;  
    l = b * aspect;  
    r = t * aspect;  
    return frustum(l, r, b, t, n, f);  
}
```

Orthographic Matrix Specification

- Convenient generation of parallel projection matrix with GLM
- `glm::ortho(left, right, bottom, top, near, far)`



- View volume does not have to be symmetrical with respect to the z -axis
- Coordinate system still right-handed



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4.5 Orthographic Projection Derivation

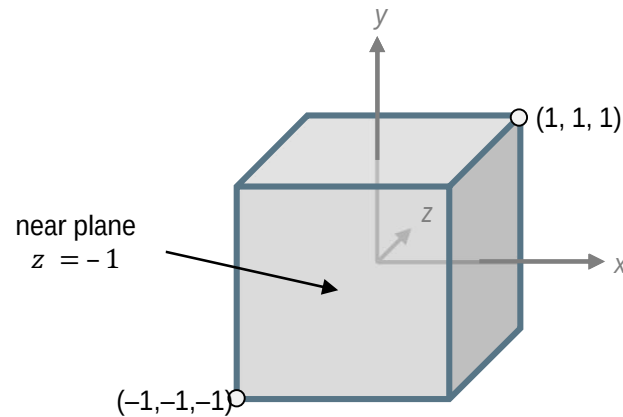
Step-wise transformation of a cuboid into a cube



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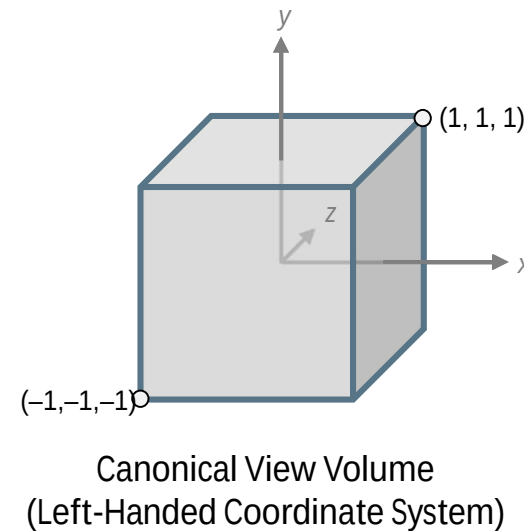
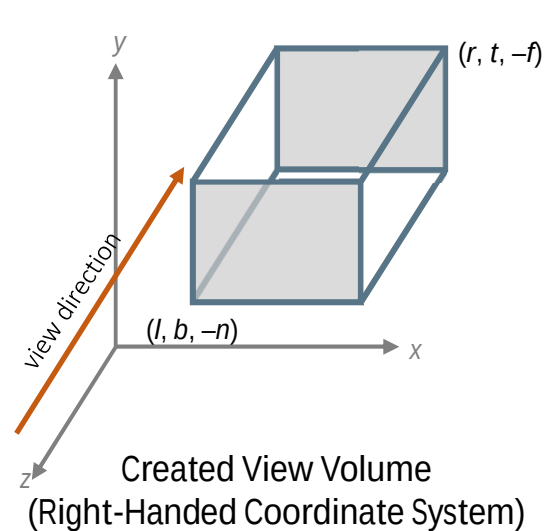
Canonical View Volume

- Canonical view volume – simplest possible view volume
 - Cube centered at the origin
 - Cube faces $x = \pm 1$, $y = \pm 1$, $z = \pm 1$
 - Defined in a left-handed coordinate system (LHS)



Orthographic Projection 1/4

- Transform right-handed view volume specified as $[l, r] \times [b, t] \times [-n, -f]$ into left-handed canonical view volume $[-1, 1] \times [-1, 1] \times [-1, 1]$ and then create the projected image by neglecting z-values



Orthographic Projection 2/4

- Let $p(x, y, z)$ be a model point and $q(x', y', z')$ its transformed point
- The desired pixel (x_p, y_p) results from orthographic projection of q onto the plane $z = -1$: $x_p = x'$ and $y_p = y'$
- Orthographic projection consists of three transformation steps
 - Translation T
 - Scaling S
 - Mirroring SP
- Step 1, translate view volume into origin $(0, 0, 0)$
 - $T = T(-\frac{r+l}{2}, -\frac{t+b}{2}, -\frac{f+n}{2})$

Orthographic Projection 3/4

- Step 2, scale cuboid from step 1 into cube

- Original cuboid bounds are

$$\left[-\frac{r-l}{2}, \frac{r-l}{2}\right] \times \left[-\frac{t-b}{2}, \frac{t-b}{2}\right] \times \left[-\frac{f-n}{2}, \frac{f-n}{2}\right]$$

- Cube bounds shall be

$$[-1,1] \times [-1,1] \times [-1,1]$$

- Resulting transformation

$$S = S\left(\frac{2}{r-l}, \frac{2}{t-b}, \frac{2}{f-n}\right)$$

- Step 3, mirror cube from step 2 to get left-handed coordinate system

$$SP = M_z = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Orthographic Projection 4/4

- Multiplication of all orthographic projection matrices results in

$$P_{orth} = SP \cdot S \cdot T = \begin{pmatrix} \frac{2}{r-l} & 0 & 0 & -\frac{r+l}{r-l} \\ 0 & \frac{2}{t-b} & 0 & -\frac{t+b}{t-b} \\ 0 & 0 & -\frac{2}{f-n} & -\frac{f+n}{f-n} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- An orthographic projection of the transformed objects followed by a hidden surface removal provides the desired image



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4.6 Perspective Projection Derivation

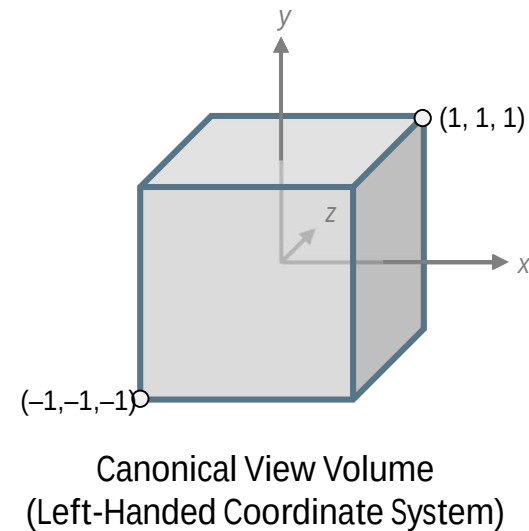
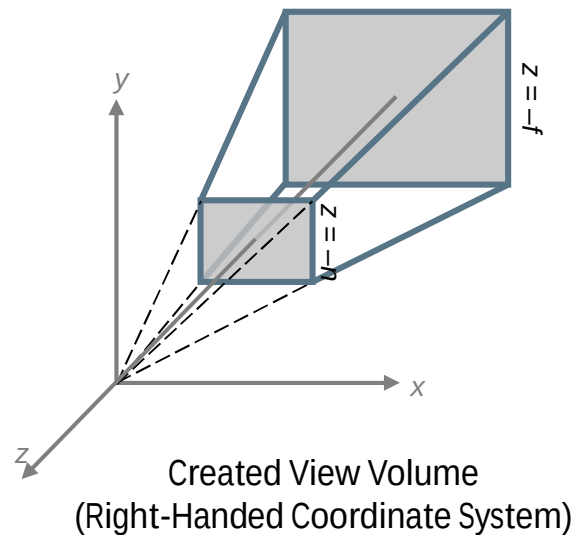
Step-wise transformation of a frustum into a cube



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Perspective Projection

- Transform view volume created by `glm::frustum` or `glm::perspective` into canonical view volume and then create the projected image by neglecting z-values



Transformation of z-Values 1/2

- After the perspective projection, z-values shall be mapped as follows
 - All z-values lying on the near plane shall have a value of -1
 - All z-values lying on the far plane shall have a value of 1
- The following matrix P_{pe} allows modification of the depth values and assures the flip (left-handed to right-handed conversion)

- $P_{pe} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & -1 & 0 \end{pmatrix}$, as

- $P_{pe} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \\ az + b \\ -z \end{pmatrix}$, whereby $-z$ is a positive number

Transformation of z-Values 2/2

- Choosing values for a and b in P_{pe}

- a and b must fulfill the following

$$P_{pe} \cdot (0 \ 0 \ -n \ 1)^T = (0 \ 0 \ -an + b \ n)^T \xrightarrow{\div w} \left(0 \ 0 \ \frac{-an + b}{n}\right)^T \hat{=} (0 \ 0 \ -1)$$

$$P_{pe} \cdot (0 \ 0 \ -f \ 1)^T = (0 \ 0 \ -af + b \ f)^T \xrightarrow{\div w} \left(0 \ 0 \ \frac{-af + b}{f}\right)^T \hat{=} (0 \ 0 \ 1)$$

- Solving the following system of linear equations

$$-an + b = -n$$

$$-af + b = f$$

- results in $a = -\frac{f+n}{f-n}$

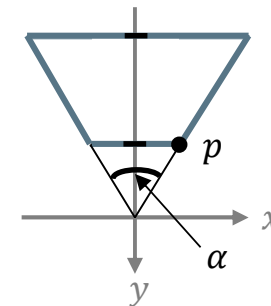
- Substituting a results in $b = -\frac{2nf}{f-n}$

Transformation of x- and y-Values 1/2

- Derivation so far leaves us with the following matrix which only changes z-coordinates

$$\bullet P_{pe} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{f+n}{f-n} & -\frac{2nf}{f-n} \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

- To transform x- and y-coordinates, we consider view frustum corner point p
 - Due to $\frac{r}{n} = \tan\left(\frac{\alpha}{2}\right)$, we can express coordinates of p as $p = (n \cdot \tan\left(\frac{\alpha}{2}\right), 0, -n)$



Transformation of x- and y-Values 2/2

- To transform p to the coordinates (1,0,-1), the following equation must hold

$$\bullet P_{pe} \cdot \begin{pmatrix} n \cdot \tan\left(\frac{\alpha}{2}\right) \\ 0 \\ -n \\ 1 \end{pmatrix} = \begin{pmatrix} n \cdot \tan\left(\frac{\alpha}{2}\right) \\ 0 \\ +n \left(\frac{f+n}{f-n}\right) - \frac{2nf}{f-n} \\ n \end{pmatrix} \xrightarrow[\text{coord. } (=n)]{\div w} \begin{pmatrix} \tan\left(\frac{\alpha}{2}\right) \\ 0 \\ \frac{+n \left(\frac{f+n}{f-n}\right) - \frac{2nf}{f-n}}{n} \\ 1 \end{pmatrix} \hat{=} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

- Thus, the perspective projection matrix is

$$\bullet P_{pe} = \begin{pmatrix} \cot\left(\frac{\alpha}{2}\right) & 0 & 0 & 0 \\ 0 & \cot\left(\frac{\alpha}{2}\right) & 0 & 0 \\ 0 & 0 & -\frac{f+n}{f-n} & -\frac{2nf}{f-n} \\ 0 & 0 & -1 & 0 \end{pmatrix}$$



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4.7 Viewport Transformation

Mapping the normalized device coordinates to the screen



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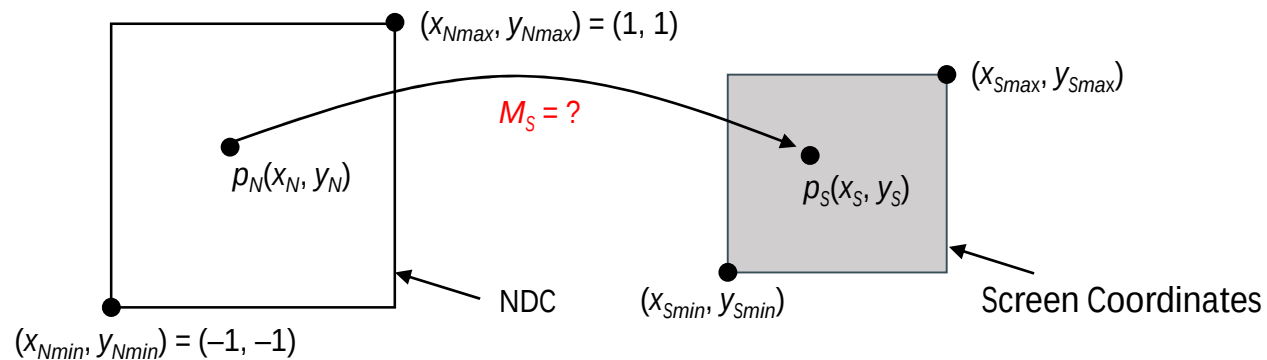
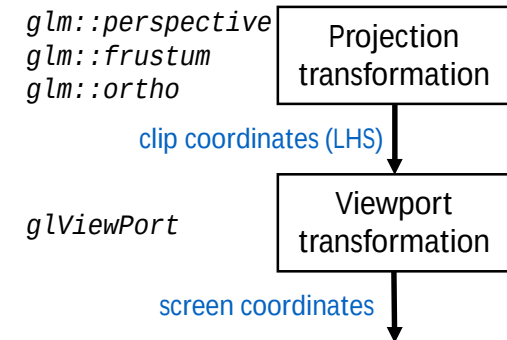
Screen Coordinate Mapping

- Mapping of the viewing plane on a drawing surface
- Define the pixel rectangle to be mapped to
 - Initially, this rectangle is the size of the entire canvas
- `glViewport(GLint x, GLint y, GLsizei width, GLsizei height)`
- When window size changes, adjust the aspect ratio to avoid distortion

```
glViewport(0, 0, 400, 200);  
glm::perspective(fovy, 2.0, near, far);
```

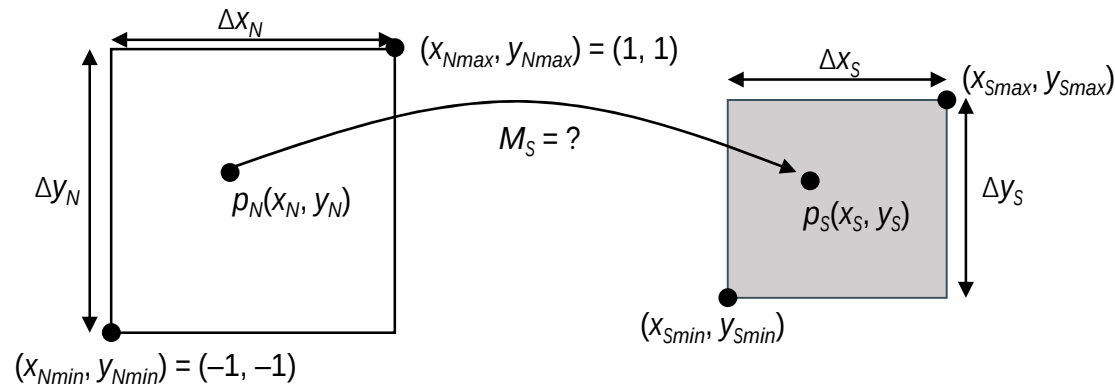
Viewport Transformation 1/2

- Specification of the viewport parameters by
- `glViewport(x, y, w, h)`



Viewport Transformation 2/2

- Viewport transformation is a 2D transformation defined as $p_s = M_s \cdot p_N$ with
 - $M_s = T(x_{Smin}, y_{Smin}) \cdot S\left(\frac{\Delta x_S}{\Delta x_N}, \frac{\Delta y_S}{\Delta y_N}\right) \cdot T(-x_{Nmin}, -y_{Nmin})$
- Thus, screen coordinates are computed as
 - $x_S = x_{Smin} + (1 + x_N) \frac{\Delta x_S}{2}; y_S = y_{Smin} + (1 + y_N) \frac{\Delta y_S}{2}$



Chapter Summary

- Projection has its origins in the arts
- Based on the deformation of the initial view volume to the canonical view volume, we can differentiate different projection types
- In OpenGL two main projection types exist
 - Perspective projection results in perspective shortening
 - Orthographic projection remains sizes
- After projection the view transformation transforms into pixel coordinates



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References

Literature and other sources used in this chapter



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- P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer Graphics (4th Edition), AK Peters 2016. (Chapter 7)